

Introduction to GPU Optimization



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1. 과제 해답

1.1. 해답

Circular Convolution!

$$y[n] = \sum_{k=0}^{N-1} h[k] \cdot x[(n - k) \bmod N]$$

실제 구현: 왼쪽/위쪽에 오른쪽/아래쪽의 픽셀을 적절하게 복붙.

1.2. 해답(심층)

DFT(Discrete Fourier Transform)

2. The Hardware(sort of)

AI 기술의 발전

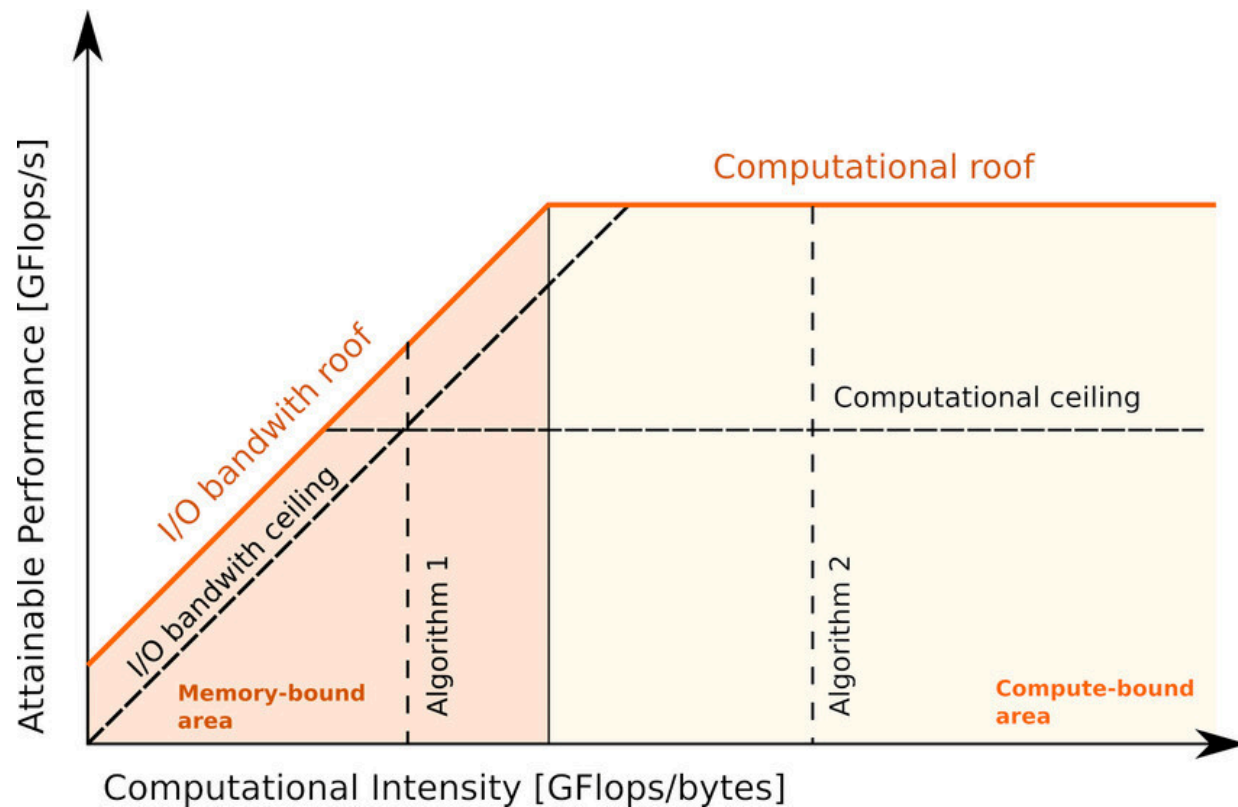
- 수학
회귀 분석, 확률론, ...
- 컴퓨터
하드웨어 가속기, 알고리즘, ...
- 둘다?
고전적 제어이론, 최신 트렌드
Mamba-3

“Dilemma” of programming

Modularity or Coupled?

하이 레벨 프로그래밍은 하드웨어가 보이지 ~~않는다~~ 않아야 한다
로우 레벨로 내려가면...

The Roofline Model



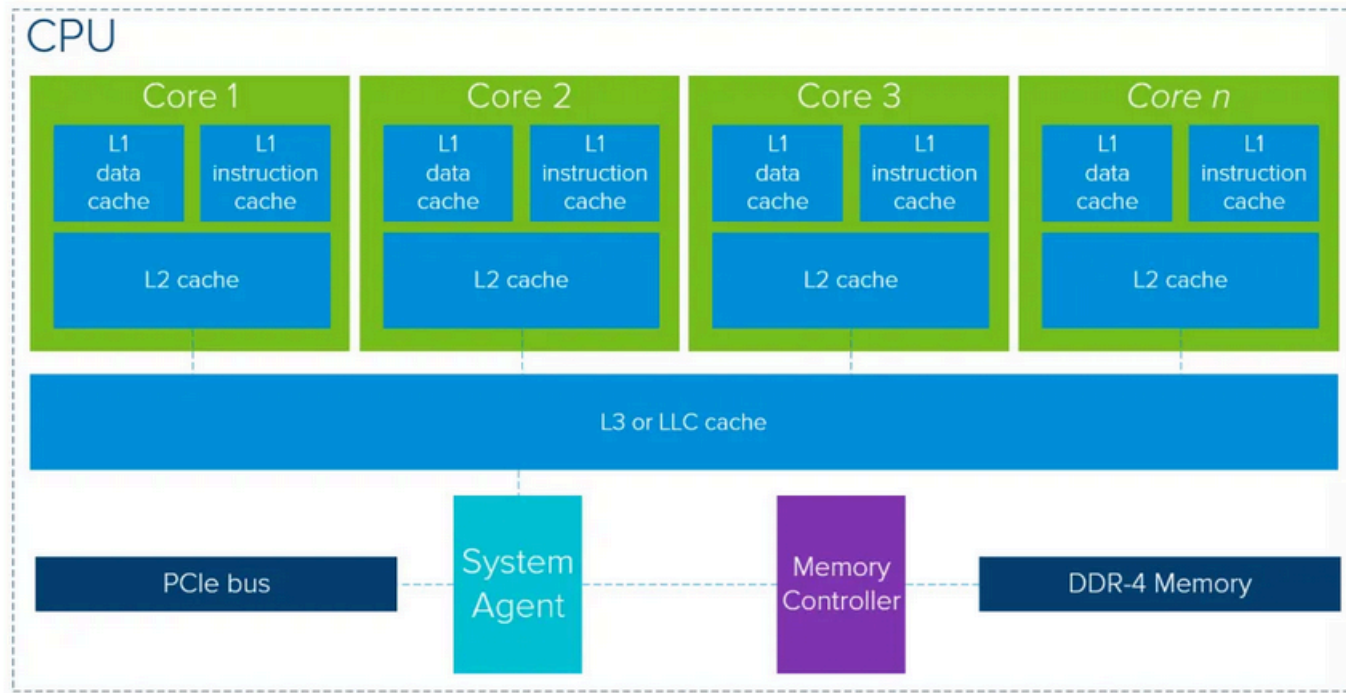
The Roofline Model

Memory-bound vs Compute-bound

How to improve positioning of ridge point?

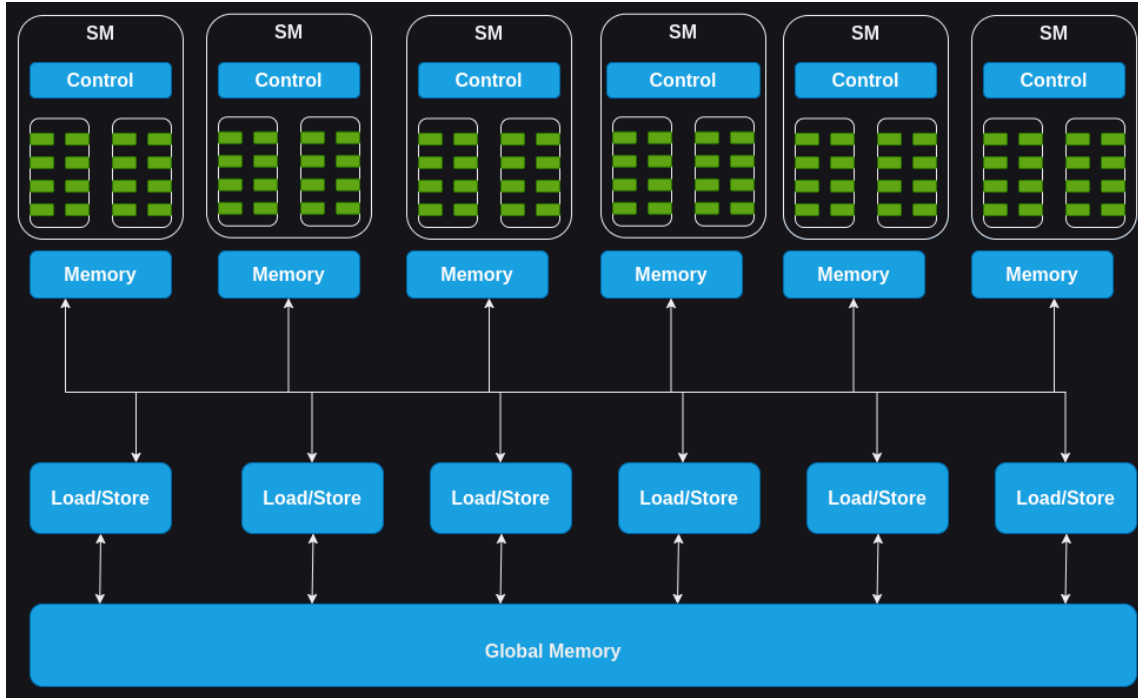
CPU vs GPU

CPU vs GPU



Core + Memory $\frac{\text{CPU}}{\text{GPU}}$.

CPU vs GPU



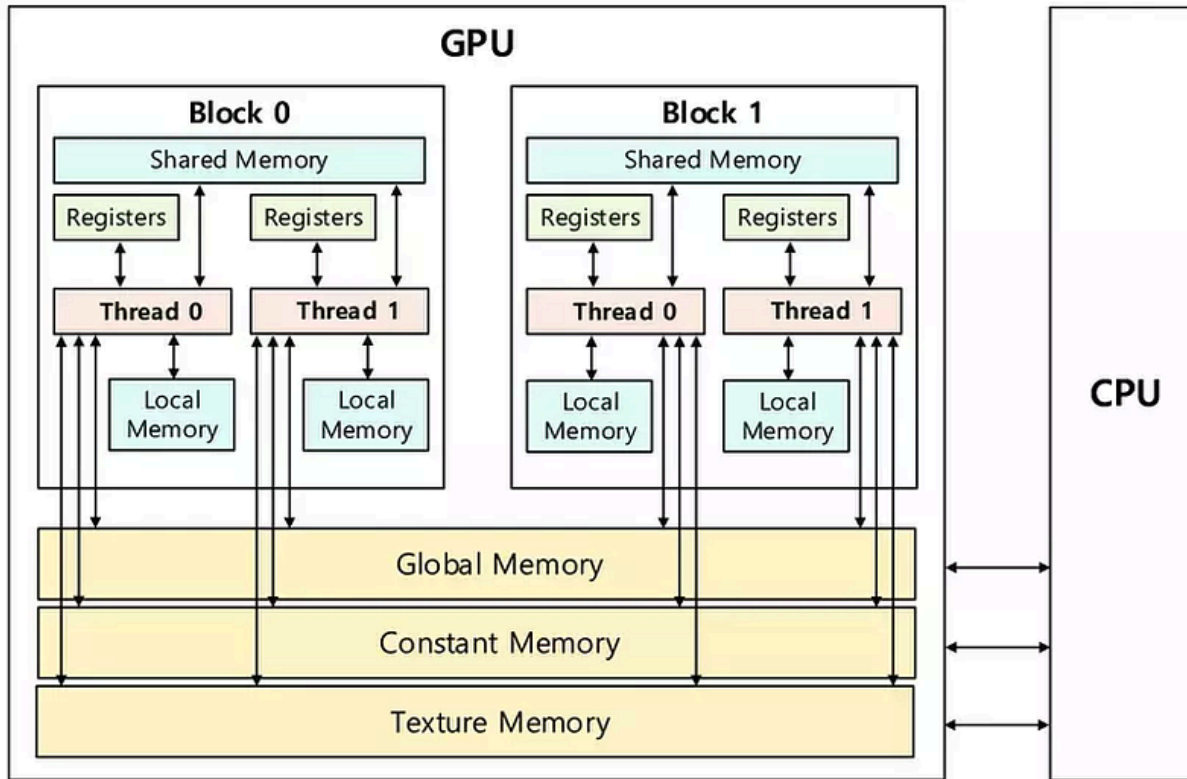
뭐가 엄청 많음;;

GPU Hierarchy

- Compute
thread < block < grid
- Memory
register < shared mem < cache < global mem

두 가지는 서로 연관되어 있음

GPU Hierarchy



3. Handling Memory-Bound

Types of Performance Bounds

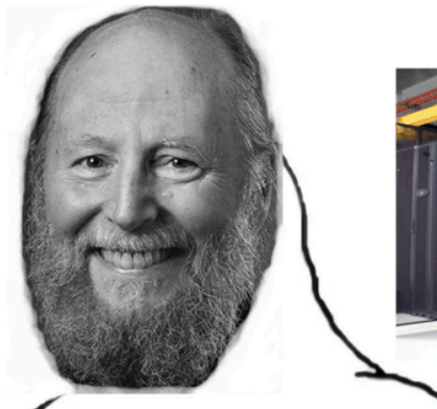
- Compute-bound
- Memory-Bound
- Overhead

어떤거가 병목인지 이해할 필요가 있음

Types of Performance Bounds



noooooo you can't just scale up pure connectionist models on Internet data without inductive biases and modularization and expect them to learn real-world knowledge and grammar from form, or arithmetic and logical reasoning and causal inference—that's just memorization and superficial pattern-matching like Eliza, you need grounding in real-world communication with intent and social dynamics and multimodal robotic embodiment which can foster disentangled learning from guided exploration and self-directed goals expressed in Bayesian programs and probabilistic graphical models which are interpretable and pin down a unique semantics which can be debiased and expressed with uncertainty, and learned efficiently on tiny academic budgets, the cost only shows how this is a dead-end, we need to stop chasing SOTAs and model the complexity of the brain and consider the social context to decolonize AI's structural biases for Third World researchers...

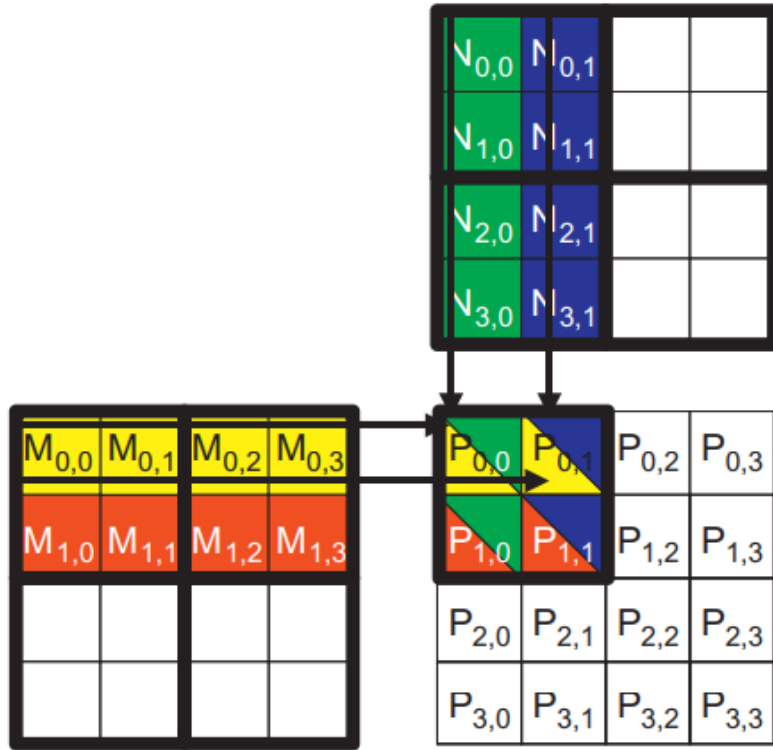


haha gpus go bitterrr



실습: CUDA

Tiling in Mat Muls



Kernel Fusion

실습: Triton